The Complete Structural Mereology of a Neurosymbolic Search-and-Rescue Manifold: A Unified Axiomatic System Specification

System Architecture Invariant Manifest

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1 Introduction: The Unified 11-Part Search-and-Rescue Plot

We formalize the search-and-rescue plot for a central Fire-Station as an interlocking multi-graph Sequential Dynamic System (SDS) operating across distinct modal boundaries. Following Edward Zalta's Theory of Abstract Objects, the spatiotemporal execution plane maps to ordinary object exemplification (Fx), while the invariant control forests encode properties atemporally (xF). The ultimate objective of the system is the minimization of coordinate uncertainty to realize Mesarovic's Attainable Space (V'), culminating in physical vehicle extraction.

2 Axiomatic Metaphysics and the Causal Tracking Loop

We formalize the system trajectory of an asset as a higher-order relational tracking loop:

$$g(m, u, v) \equiv \exists y[R_{\text{system}}(m, u, y) \wedge E(m, u, y, v)] \tag{1}$$

where u represents environmental Navier-Stokes advection forces, m represents the external divine control input vector, y represents human or machine output behaviors, and E represents the absolute spiritual evaluation function.

3 The 27-Dimensional Integrated Attribute Matrix

To inject continuous meaning and speech-act context into the message-passing convolutions, the GNN ingests an integrated, multi-scale node attribute matrix $\boldsymbol{X} \in \mathbb{R}^{N \times 27}$. For each node i, the vector decomposes into three distinct socio-pragmatic tensor blocks:

$$\boldsymbol{x}_i = \left[\boldsymbol{e}_i^{\text{empirical}} \mid \boldsymbol{c}_i^{\text{cognitive}} \mid \boldsymbol{s}_i^{\text{institutional}} \right] \tag{2}$$

where $\boldsymbol{e}_i \in \mathbb{R}^{18}$ encapsulates the Barna loop splits, the 72% Pew United States Church macro-advection wavelets, and the JAX spatial grid coordinates. The cognitive component $\boldsymbol{c}_i \in \mathbb{R}^6$ packs the compiled LISP bar preferences and the 3-dimensional Rational Speech Act (RSA) scalar implicature distributions. The institutional component $\boldsymbol{s}_i \in \mathbb{R}^3$ isolates the zero-uncertainty Searlean speech-act flags (X counts as Y in C), allowing the network to collapse giant graph topologies down to a unified virtual 0-simplex token ready for decision tree lookup routing.

4 Bayesian Belief Polarization and Worldview Channels

We formalize the divergence profiles of Example 2 (Religious Belief) and Example 4 (Political Policy) as a normative Bayesian update process operating across decoupled background knowledge tensors.

4.1 The Religious Belief Update Formulation

Let $H \in \{0, 1\}$ be the divinity hypothesis space, $V \in \{0, 1\}$ be the background universe context (Secular vs. Christian), and $D \in \{0, 1\}$ represent the observed data vector (Persecution). The prior probability of divinity is explicitly conditioned on the worldview parameter: $P(H \mid V)$. The final updated posterior distribution $P(H = 1 \mid D = 1)$ is computed via:

$$P(H = 1 \mid D = 1) = \frac{\sum_{v \in \{0,1\}} P(D = 1 \mid H = 1, v) P(H = 1 \mid v) P(v)}{\sum_{h \in \{0,1\}} \sum_{v \in \{0,1\}} P(D = 1 \mid h, v) P(h \mid v) P(v)} \tag{3}$$

Because Alice assigns $P(V = 1) = 0.9$ and Bob assigns $P(V = 1) = 0.1$, the identical observation vector $D = 1$ updates Alice's belief upward to 0.8604 while causing Bob's posterior to experience a downward advection trend, demonstrating absolute cognitive polarization under perfect rationality.

4.2 The Political Policy Update Formulation

Let $H \in \{0, 1\}$ be the policy evaluation space (Bad vs. Good), $V_1 \in \{0, 1\}$ represent the agent's background economic philosophy (Conservative vs. Liberal), and $V_2 \in \{0, 1\}$ represent the bill's spending outcome. The independent study output matches $D \in \{0, 1\}$. Evaluated over uniform priors for V_2, the strategic posterior joint matrix unrolls via:

$$P(H = 1 \mid D = 1) = \frac{\sum_{v_1} \sum_{v_2} P(H = 1 \mid v_1, v_2) P(D = 1 \mid v_2) P(v_1) P(v_2)}{\sum_{h} \sum_{v_1} \sum_{v_2} P(h \mid v_1, v_2) P(D = 1 \mid v_2) P(v_1) P(v_2)} \tag{4}$$

Because Alice attaches a 0.90 prior to $V_1 = 1$, her belief surges from 0.6600 up to 0.7880 upon receiving an incriminating spending signal ($D = 1$). Bob's prior maps inversely, driving his posterior down to 0.2120. This provides the exact mathematical proof that a single study dataset expands belief variance across polarized network clusters.